# Alternative construction of $\mathcal{U}\otimes V$.

Let $S$ be a non-empty set. Consider the set

$F_0(S,K) := \{ f: S \to K \mid \#\{s\in S \mid f(s)\neq 0\} < \infty \}$ of functions

$S \to K$ with finite support. Note that $F_0(S,K)$ is a vect. sp.

and in fact $\exists$ a canonical iso. $q: F_0(S,K) \xrightarrow{\cong} FoSp_K(S)$.

$$\left(\begin{array}{ll} \text{Indeed} & F_0(S,K) \ni f \xmapsto{q} \sum_{s\in S} f(s)\cdot \hat{s} \in FoSp_K(S) \\ & \quad\quad \rotatebox{90}{$\in$} \qquad\qquad\qquad\qquad \rotatebox{90}{$\in$} \\ \text{and} & \left( f(s) := a_s, \forall s\in S \right) \xleftarrow{q^{-1}} \sum_{s\in S} a_s \cdot \hat{s} \end{array}\right)$$

If $S=\phi$ we define $F_0(S,K) := 0$ the $0$-vect.sp.

Let $s_0 \in S$ and denote by $\delta_{s_0} \in F_0(S,K)$ the function

$$\delta_{s_0}(s) := \begin{cases} 1 & s=s_0 \\ 0 & s\neq s_0 \end{cases}.$$

<u>Exc.</u> The set $\{\delta_{s_0} \mid s_0 \in S\}$ forms a basis for $F_0(S,K)$.

<u>Exc.</u> Let $\mathcal{U}$ be a vect. sp. over $K$ with a basis $S$.

Then $\exists$ a canonical iso. (that depends on $S$),

$$F_0(S,K) \xrightarrow{\cong} \mathcal{U}. \qquad f \longmapsto \sum_{s\in S} f(s)\cdot s.$$

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So let's assume that $\mathcal{U} = F_0(S,K)$, $V = F_0(Q,K)$ for

some sets $S$ & $Q$.

Define $X := F_0(S\times Q) = \left\{ f: S\times Q \to K \;\middle|\; \#\{(s,q) \mid f(s,q)\neq 0\} < \infty \right\}$.

Define also $\tau: \mathcal{U}\times V \to X$,

$\tau(u,v) :=$ the function $S\times Q \to K$, $(s,q) \mapsto u(s)v(q)$

(i.e. $\tau(u,v): S\times Q \to K$, $(\tau(u,v))(s,q) = u(s)v(q)$.)

Exc. 1) $\tau(u,v)$ is indeed in $F_0(S\times Q)$.

2) $\tau$ is bilinear.

3) $\forall\, s_0\in S,\ q_0\in Q,\ \ \tau(\delta_{s_0},\delta_{q_0})=\delta_{(s_0,q_0)}\in X$

We claim that $(X,\tau)$ is a tensor product of $\mathcal{U}$ & $V$.

To see this, let $W$ be any vect. sp. over $K$ and $\Psi:\mathcal{U}\times V\longrightarrow W$ any bilinear map. Define $\varphi: \underset{F_0(S\times Q,K)}{\overset{\shortparallel}{X}}\longrightarrow W$ to be the unique linear map s.t. $\varphi(\delta_{(s_0,q_0)}):=\Psi(\delta_{s_0},\delta_{q_0})\in W$.

Since $\{\delta_{(s_0,q_0)}\}_{\substack{s_0\in S\\ q_0\in Q}}$ is a basis for $X=F_0(S\times Q,K)$, $\varphi$ is a well defined linear map.

We claim that $\varphi\circ\tau=\Psi$. To see this, let $u\in\mathcal{U}, v\in V$.

We have $u=\sum_{s_0\in S}u(s_0)\cdot\delta_{s_0}$, $v=\sum_{q_0\in Q}v(q_0)\cdot\delta_{q_0}$.

$$\Rightarrow \Psi(u,v)\underset{\Psi\text{ is bilin.}}{=}\sum_{s_0\in S}\sum_{q_0\in Q}u(s_0)\cdot v(q_0)\cdot\underbrace{\Psi(\delta_{s_0},\delta_{q_0})}_{\varphi(\delta_{(s_0,q_0)})}=$$

$$\underset{\varphi\text{ is lin.}}{=}\varphi\Big(\sum_{s_0\in S,q_0\in Q}u(s_0)\cdot v(q_0)\cdot\delta_{(s_0,q_0)}\Big)=$$

$$=\varphi\Big(\tau\big(\sum_{s_0\in S}u(s_0)\delta_{s_0},\ \sum_{q_0\in Q}v(q_0)\delta_{q_0}\big)\Big)=\varphi\circ\tau(u,v).$$

Finally, $\varphi$ is unique. This follows from the fact that $\operatorname{Im}\tau$ spans $X$ like in the 1'st proof (indeed $\forall\, s_0,q_0$, $\delta_{(s_0,q_0)}$ is in the image of $\tau$, and $\{\delta_{(s_0,q_0)}\}_{\substack{s_0\in S\\ q_0\in Q}}$ span $X$). ∎

So, in the model of this construction $\delta_{s_0} \otimes \delta_{q_0} = \delta_{(s_0, q_0)}$.

A quick Cor. of the alternative construction.

<u>Cor.</u> If $S$ is a basis for $\mathcal{U}$ and $Q$ is a basis for $V$, then $\{s \otimes q \mid s \in S, q \in Q\}$ is a basis for $\mathcal{U} \otimes V$ and this basis is in 1-1 correspondence with the set $S \times Q$. In particular, if $\mathcal{U}$ & $V$ are f. dim. then $\dim \mathcal{U} \otimes V = (\dim \mathcal{U}) \cdot (\dim V)$.

E.g. $K^m \otimes K^n \cong K^{m \cdot n}$.